

Morphology Beats Multilingual Embeddings for Kazakh Retrieval: A 300-Query Benchmark with Honest Negative Results

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Abstract

We present a reproducible retrieval benchmark for Kazakh — an agglutinative, low-resource language — comprising 300 queries over 8 370 Wikipedia passages across three query categories: inflected (morphological stress-test), natural, and vocabulary-gap (synonym stress-test). We evaluate five retrieval systems: BM25 with and without a Kazakh stemmer, and three zero-shot dense models (LaBSE, Granite-278M, E5-base). The Kazakh stemmer significantly improves BM25 retrieval (+16% nDCG@10 on inflected queries, $p=0.0017$; +9% overall, $p=0.0001$, $n=300$, paired bootstrap). BM25 with the stemmer outperforms zero-shot LaBSE (0.754 vs 0.481 overall). We also report three honest negative results: (1) naive unweighted synonym query expansion hurts all categories (-0.215 nDCG@10); (2) RRF fusion of BM25+Stemmer and Granite fails both pre-registered success criteria; (3) better retrieval does not produce a significant end-to-end RAG accuracy gain on Qwen2.5-7B (McNemar $p=0.63$). All code, data, and results are publicly available.

1. Introduction

Kazakh is a Turkic agglutinative language spoken by approximately 13 million people, primarily in Kazakhstan. Its morphology is exceptionally productive: a single root can generate dozens of surface forms through the sequential attachment of suffixes encoding case, number, possession, tense, and other grammatical categories. A word such as *мeңiз* (sea) appears in text as *мeңiздеpze* (to the seas), *мeңiзде* (at the sea), or *мeңiздiң* (of the sea) — all sharing the same root but differing in surface form. For lexical retrieval systems that rely on exact or near-exact token matching, this morphological diversity is a direct failure mode: a query containing *мeңiз* will not match a passage containing *мeңiздеpze* unless some form of normalization is applied. In our corpus of 800 Wikipedia articles, we observe **102 408 unique surface forms** — a scale of lexical fragmentation that makes morphological normalization a first-order concern for any Kazakh information retrieval system.

Despite Kazakh's significance as a national language and growing digital footprint, it remains severely underrepresented in information retrieval research. To our knowledge, no prior work has presented a statistically validated, reproducible IR benchmark for Kazakh that (a) covers multiple query types, (b) compares sparse and dense retrieval systems on the same query set, and (c) reports significance tests on the results. Practitioners building Kazakh search or RAG systems must currently rely on intuition, multilingual model documentation, or benchmarks from morphologically simpler languages — none of which adequately predict behavior on agglutinative input. We address this gap directly.

In this paper we present a 300-query retrieval benchmark over 8 370 Kazakh Wikipedia passages. We evaluate five systems — BM25 with and without a morphological stemmer, and three zero-shot multilingual dense models — and report results with paired bootstrap significance tests. Our central finding is that a Kazakh stemmer (+9% nDCG@10 overall, $p=0.0001$) outperforms zero-shot LaBSE embeddings (0.754 vs 0.481), establishing that morphological normalization is more valuable than naive multilingual vectorization for this language. We also report three pre-registered negative results — synonym query expansion, RRF hybrid fusion, and end-to-end RAG accuracy — which we include in full, as negative results clarify the remaining open problems more honestly than selectively reporting only what worked.

2. Related Work

Morphological normalization in lexical IR. The effect of stemming and lemmatization on retrieval quality is well established for morphologically rich languages. Stemming yields substantial gains for highly inflected and agglutinative languages such as Finnish, Hungarian, and Turkish, while offering little benefit for morphologically simple languages like English (Robertson & Zaragoza, 2009). Kazakh, as a Turkic agglutinative language, sits firmly in the high-benefit category, yet no published benchmark quantifies this effect on Kazakh retrieval with statistical validation — the gap this work addresses.

Multilingual dense retrieval. Multilingual sentence encoders such as LaBSE (Feng et al., 2022), multilingual-E5 (Wang et al., 2024), and IBM Granite embeddings (IBM Research, 2024) advertise coverage of 100+ languages and strong cross-lingual retrieval. However, headline benchmarks are dominated by high-resource languages, and zero-shot performance on low-resource agglutinative languages is rarely reported in isolation. We evaluate three such models head-to-head against a lexical baseline on Kazakh, and find that morphological normalization on a sparse index outperforms zero-shot LaBSE — a result that cautions against assuming multilingual embeddings transfer cleanly to this setting.

Query expansion. Expanding queries with synonyms or related terms is a classic technique for closing the vocabulary gap between queries and documents (Voorhees, 1994). Its effect is known to depend heavily on the strength of the underlying retrieval signal: when the lexical match is already strong, unweighted expansion can introduce more noise than signal (Robertson & Zaragoza, 2009). We confirm this empirically for Kazakh and provide a fine-grained subgroup analysis isolating the failure mechanism.

Retrieval-augmented generation. RAG pipelines couple a retriever with a generative LLM. A common implicit assumption is that improving retrieval improves end-to-end answer quality. We test this assumption directly on Kazakh and find that a significant retrieval improvement does not yield a significant end-to-end accuracy gain on a 7B-parameter generator — localizing the bottleneck to generation rather than retrieval.

3. Data & Methodology

3.1 Corpus

800 random Kazakh Wikipedia articles from the `wikimedia/wikipedia` dataset (Hugging Face) were cleaned, split into ~120-word passages, and filtered for Kazakh language content, yielding **8 370 passages**. The

corpus contains **102 408 unique surface forms** — illustrating the morphological diversity that motivates this work.

3.2 Queries

We constructed **300 queries** covering 100 named entities (countries, cities, people, and concepts) $\times$ 3 categories:

- **inflected** (100 queries): the key word appears in an oblique grammatical case (e.g., genitive, locative). Tests morphological robustness.
- **natural** (100 queries): standard factual questions. Baseline category.
- **vocabulary-gap** (100 queries): the query uses a synonym or paraphrase of the word used in the gold passage. Tests semantic bridging.

Each query has one ground-truth passage (qrels). Queries are short factual questions with a one-to-two word answer (e.g., "Қазақстанның астанасы қайда?" $\rightarrow$ "Астана").

3.3 Systems

#	System	Description
1	BM25	Okapi BM25 ($k_1=1.5$, $b=0.75$), no normalization
2	BM25 + Stemmer	Same BM25, corpus and queries stemmed with Kazakh stemmer
3	Dense LaBSE	sentence-transformers/LaBSE , zero-shot
4	Dense Granite	ibm-granite/granite-embedding-278m-multilingual , zero-shot
5	Dense E5	intfloat/multilingual-e5-base , zero-shot

Dense models evaluated zero-shot (no fine-tuning, no hard-negative training). Similarity: cosine (L2-normalized $\rightarrow$ dot product). Brute-force exact search, no FAISS. Input formats per model card: E5 with query: / passage: prefixes; LaBSE and Granite without prefixes (symmetric models).

3.4 Metrics

Recall@{1,5,10}, MRR@10, nDCG@10. Statistical significance: paired bootstrap, 10 000 resamples, two-sided.

3.5 RAG Evaluation

For end-to-end evaluation, we ran BM25 (with and without stemmer) $\rightarrow$ top-3 passages $\rightarrow$ Qwen2.5-7B (4-bit quantization) on all 300 queries. Accuracy measured by substring match: **correct** if the gold answer string appears in the response, **abstain** if the model wrote "Ақпарат жоқ", **hallucination** otherwise. This is conservative — a semantically correct but differently phrased answer counts as hallucination. Paired significance: McNemar exact test.

4. Results

4.1 Main Retrieval Results (nDCG@10, n=300)

System	inflected	natural	vocab-gap	ALL
BM25	0.627	0.703	0.741	0.690
BM25 + Stemmer	0.727	0.772	0.764	0.754
Dense LaBSE	0.477	0.546	0.419	0.481
Dense Granite	0.791	0.923	0.303	0.672
Dense E5	0.845	0.947	0.562	0.785

4.2 Statistical Significance (BM25 → BM25 + Stemmer)

category	Before	After	Δ	gain	p-value
inflected	0.627	0.727	+0.101	+16%	p=0.0017
natural	0.703	0.772	+0.068	+10%	p=0.0063
vocabulary-gap	0.741	0.764	+0.023	+3%	p=0.21 ✗
ALL	0.690	0.754	+0.064	+9%	p=0.0001

The stemmer significantly improves inflected and natural queries but does **not** significantly improve vocabulary-gap (p=0.21) — the theoretically expected result.

4.3 RAG End-to-End (Qwen2.5-7B, n=300)

Retriever	hit@3	Accuracy	Hallucination	Abstain
BM25 (no stemmer)	0.737	0.483	0.217	0.300
BM25 + Stemmer	0.803	0.500	0.240	0.260
Δ	+0.066	+0.017	+0.023	-0.040

McNemar exact test: p=0.625 — not significant. The retrieval gain (+0.066 hit@3) does not translate into a significant accuracy gain. The stemmer shifts the model out of abstention (0.300 → 0.260), but recovered answers split between correct and hallucinated.

5. Negative Results

5.1 Synonym Query Expansion Hurts Retrieval

We tested unweighted synonym query expansion: for each query stem, synonyms from a Kazakh synonym dictionary (736 query stems; 30% coverage) were appended to the query. Queries expanded from ~6.5 to ~26.5 tokens on average.

System	inflected	natural	vocab-gap	ALL
BM25 + Stemmer	0.727	0.772	0.764	0.754
BM25 + Synonym Expansion	0.537	0.451	0.627	0.539
Δ	-0.190	-0.321	-0.137	-0.215

To diagnose the failure, we split vocabulary-gap queries into three subgroups:

Subgroup	n	Stemmer nDCG@10	+Synonym nDCG@10	Δ
uncovered (no synonyms)	2 (2%)	1.000	1.000	≈ 0
covered_noise (synonyms $\nsubseteq$ gold)	73 (73%)	0.751	0.570	-0.181
covered_bridge (synonyms $\cap$ gold $\neq \emptyset$)	25 (25%)	0.783	0.766	-0.017

Even `covered_bridge` — where the correct synonym was added — drops slightly (-0.017). The dominant failure mode is `covered_noise` (73%): contextually wrong synonyms bloat the query and pull spurious documents. The expansion mechanism itself is the problem, not dictionary coverage.

5.2 Hybrid RRF Fails Pre-registered Criteria

We fused BM25+Stemmer and Granite rankings via RRF ($k=60$, fixed before the experiment). Pre-registered success criteria: (1) $\text{nDCG@10}(\text{Hybrid}) \geq \max(\text{BM25}, \text{Granite})$ on ALL; (2) $\text{vocab-gap}(\text{Hybrid}) \geq \text{vocab-gap}(\text{BM25+Stemmer})$.

System	inflected	natural	vocab-gap	ALL
BM25 + Stemmer	0.727	0.772	0.764	0.754
Granite	0.791	0.923	0.303	0.672
Hybrid (RRF, $k=60$)	0.824	0.877	0.525	0.742

Both criteria fail: ALL $0.742 < 0.754$ ($p=0.27$), vocab-gap $0.525 \ll 0.764$ ($p<0.0001$). Granite's collapse on vocabulary-gap (0.303) leaks through the fusion.

5.3 RAG Replication Failure ($n=60 \rightarrow n=300$)

An earlier exploratory run ($n=60$) showed a Qwen accuracy gain of +0.100 that "scales with model competence." At $n=300$, this signal did not replicate (+0.017, $p=0.63$). The retrieval result survived the scale-up; the end-to-end RAG claim did not.

6. Discussion

Why morphology matters more than multilingual embeddings for Kazakh. 800 Wikipedia articles produce 102 408 unique surface forms. BM25 on inflected queries ($\text{nDCG@10} = 0.627$) significantly underperforms on natural queries (0.703, $p=0.0017$), confirming that morphological mismatch is a measurable, quantified problem. The stemmer resolves this by reducing surface forms to roots — a cheap, interpretable, network-free intervention that outperforms LaBSE (0.754 vs 0.481) on every category.

Granite's vocabulary-gap collapse. Granite leads on inflected (0.791) and natural (0.923) queries but collapses on vocabulary-gap (0.303). This suggests the model encodes morphological variants well but fails to bridge synonym/paraphrase gaps in Kazakh — possibly due to limited Kazakh-specific training data for this semantic relationship.

The generator bottleneck. Better retrieval (hit@3 0.737 → 0.803) does not produce a significant accuracy gain ($p=0.63$). Qwen2.5-7B pushes out of abstention but partially replaces correct abstentions with hallucinations. The bottleneck has shifted from retrieval to generation.

7. Limitations

- **Single corpus domain.** Only Kazakh Wikipedia. Results may differ on news, legal, or e-commerce text.
 - **Single LLM for RAG.** Only Qwen2.5-7B (4-bit). A stronger or Kazakh-specific generator might convert the retrieval gain into accuracy.
 - **Substring-match scoring.** Conservative: semantically correct but differently phrased answers count as hallucinations.
 - **Stemmer vs lemmatizer.** No public Kazakh lemmatizer exists for direct comparison. The stemmer performs morphological analysis to base form, functionally equivalent for retrieval, but the distinction is acknowledged.
 - **Query construction.** Queries were constructed by the author, not crowd-sourced.
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8. Conclusion

Kazakh morphology measurably breaks lexical search, and a stemmer measurably fixes it (+9% nDCG@10 overall, $p=0.0001$, $n=300$). This outperforms zero-shot LaBSE. Dense E5 is the best overall system but requires GPU. We report three honest negative results — synonym expansion, hybrid RRF, and RAG end-to-end gain — that clarify where the remaining bottlenecks lie: semantic bridging (vocabulary-gap, unsolved by any system tested) and the Kazakh generator. All code and data are publicly available at <https://github.com/Tim2190/Kaz-RAG-search-benchmark>.

Conflict of Interest

The author is the developer of the Kazakh stemmer evaluated in this benchmark (System 2, <https://qaz-api.vercel.app/>). To mitigate bias, all systems were evaluated on identical queries, corpus, and metrics; all code and data are publicly available and independently reproducible; and results unfavorable to the author's system — its statistical insignificance on vocabulary-gap ($p=0.21$), and E5's superior overall performance (0.785 vs 0.754) — are reported in full.

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